

A Novel Time-to-Amplitude Converter and a Low-Cost Wide Dynamic Range FPGA TDC for LiDAR Application

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Abstract—This article demonstrates a low-cost wide dynamic range field-programmable gate array (FPGA) time-to-digital converter (TDC) and a time-to-amplitude converter (TAC) based on an embedded XOR gate of the FPGA and a low-complexity circuitry. TACs and TDCs are important for distance measurements in light detection and ranging (LiDAR) systems in a wide range of applications from self-driving cars to topographic mapping. LiDAR's high cost, relatively large size, and complexity are important implementation challenges that need to be addressed. In the literature, external circuits are used to provide low phase noise clock signals into the FPGA TDC to measure long-range time intervals adding to the complexity and cost. Our TDC can measure a wide dynamic range of 1 ms with high precision using a low-cost FPGA board. Calibration using dynamic phase-shifting (DPS) is conducted with a resolution of 18 ps. The performance measurements gave us insight on the relative temperature insensitive regions for operating the TDC. The ideal range was found to be 24 °C–39 °C. We also present a novel TAC design to demonstrate that one XOR gate of a low-cost Artix-7 FPGA and a simple low-cost circuitry can detect a phase difference as low as 18 ps. The TAC's linearity is presented through integral nonlinearity (INL) and differential nonlinearity (DNL) test results. For every additional channel, our TAC requires only one XOR gate in the FPGA chip and a low complexity circuitry. This allows for multichannel operation and larger area availability for use in other implementations. Performance measurements show our TAC to be highly insensitive to temperature and voltage changes.

Index Terms—Dynamic phase-shifting (DPS), field-programmable gate array (FPGA), integral nonlinearity (INL) and differential nonlinearity (DNL) tests, light detection and ranging (LiDAR), time-to-amplitude converter (TAC), time-to-digital converter (TDC).

I. INTRODUCTION

TIME-TO-DIGITAL converters (TDCs) and time-to-amplitude converters (TACs) are very popular and widely chosen systems for timing critical applications. For instance, they are used in light detection and ranging (LiDAR), radio detection and ranging (RADAR), very-high-accuracy single photon avalanche diode cameras [1], state-of-the-art time-of-flight (TOF) measurements [2] as a promising solution

for the vital time measurement part of Positron emission topography [3], time correlation-based single photon counting systems [4], linear accelerators for trigger timing distribution systems [5], and fluorescence lifetime spectroscopy [6], among many other application areas.

One of the main goals of this article is to encourage researchers of limited budgets to build systems with critical time measurements, like TOF measurements in LiDAR systems, at a reduced cost. The presented TDC in this article is implemented on a low-cost general purpose field-programmable gate array (FPGA) board, and it is shown by the experiments that it can measure a wide dynamic range with a high and stable precision. To attain a time interval (TI) measurement in a wide dynamic range, multiphase clock FPGA TDCs are widely used. For instance, in [5], four 90° phase-shifted clocks in Spartan-6 FPGA are used and the achieved resolution is 1 ns with a dynamic range of 20 ms. Other wide dynamic range examples can be seen in [7], where a 625-ps resolution is attained with Virtex-5 FPGA, as well as in [8], where a 1190-ps resolution is achieved with Altera Cyclone III FPGA. However, the resolution of the multiphase clock TDCs are still much lower than other FPGA TDCs that use fine time measurement methods like tapped delay lines (TDLs), vernier delay lines, and ring oscillators. Delay lines based on the propagation delay of logic gates, like multiplexers used in fast carry look ahead adders in modern FPGAs, are widely used to attain a resolution below 100 ps [9], [10], [11], [12], [13], [14]. In [15], a TDL-based FPGA TDC is proposed with an LSB of 1.7 ps and a precision of 4.2 ps for a measurement range of 6.25 ns. In [16], the LSB of the TDL-based FPGA TDC is 10 ps while the precision is 12.83 ps. In this case, the measurement range is limited to 20 ns. In the literature, effective methods exist to achieve longer range TI measurements. For instance, in [17], an integrated time digitizer is implemented based on a new conversion method that applies multiedge coding in independent coding lines. In this work, a 2-s dynamic range is achieved with a resolution of 902 fs using a custom-designed FPGA board. The FPGA TDC in [17] is tested up to 10 ms when the clock source is a temperature-compensated crystal oscillator (TCXO), and an approximately 27-ps standard deviation is attained at 10 ms. The FPGA TDC in [17] is also tested up to 1 s when an external rubidium standard FS725 (SRS)-based clock is used and an approximately 28-ps standard deviation is achieved at 1 s. An important method to achieve a wide dynamic range

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with high resolution and precision is the Nutt interpolation method which will be explained in Section II in more detail. For instance, in [13], an FPGA TDC is designed, and a 19.6-ps single-shot precision is achieved for the 40-ns TI and a 320-ps single-shot precision is achieved for the 327.7- μ s TI. In this article, we use the Nutt interpolation method along with the carry chain-based TDL to attain a wide dynamic range with high resolution and precision.

Another main goal of this article is to show that the dynamic phase-shifting (DPS) features of the Artix-7 FPGA can be used to calibrate the FPGA TDC effectively. In this article, we present a DPS calibration resolution of 18 ps. Some of the advantages of our DPS calibration method are the very low usage of chip area as well as the use of the FPGA chip alone without the need of any external circuitry. Another advantage lies in its independence from sufficiently distributed random TIs like code density tests. Instead, the logic behind DPS calibration is not complex and depends only on increasing the phase shift between two calibration clocks.

Additional care must be given when the resolution of the DPS is increased to 18 ps. In this case, the jitter of the delay taps that are responsible for phase increments becomes comparable to the phase increment resolution. To eliminate the jitter effect as much as possible, we have written a very high speed integrated circuit (VHSIC) hardware description language (VHDL) architecture to carry out 1024 measurements that are averaged for every 18-ps phase increment. By this way, our calibration results were successful in achieving high precision over a wide dynamic range.

In 2021, Charlot *et al.* [18] presented a DPS calibration with a resolution of 78 ps which is not detailed enough to calibrate high-resolution TDCs. In 2017, Stofanik *et al.* [19] presented a DPS technique with 40-ps resolution in a frequency-controlled clock system that is not related to the calibration of the TDCs. In 2016, Vornicu *et al.* [20] presented a time interval generator (TIG) with 27-ps resolution for use in the calibration of TDCs. It has been proven in [20] that a 27-ps TI increment resolution in DPS calibration is sufficient in effectively calibrating the TDCs by comparing the DPS calibration with classical code density test and then subsequently comparing the calibrated TDC results with state-of-the-art PicoHarp 300 time-correlated single photon counting (TCSPC) system. In this article, we conduct calibration using DPS with resolution as low as 18-ps. Our experimental results show the effectiveness of our high-resolution DPS calibration technique in calibrating our FPGA TDC.

In this article, we present an FPGA TDC to show that a wide dynamic range can be measured with high precision using a very low-cost, general-purpose FPGA board. Our experiments showed that the precision of the TI measurement of our FPGA TDC does not significantly change with the TI range, thus displaying a stable behavior. The mean of the standard deviation up to 1 μ s is 33.9 ps while the mean of the standard deviation from 1 μ s up to 1 ms is 41 ps. This is promising for applications where measuring long TI range is important like in LiDARs, timing distribution systems, and RADARs.

The propagation delay of carry cells is affected by the process, voltage, and temperature (PVT) variations. Hence, we carried out performance analysis to measure the effect of

the temperature change read by the analog-to-digital converter of AMD Xilinx (XADC) of the FPGA on the propagation delay of carry cells. The temperature was varied from -30°C to 105°C . Our experimental results showed a great temperature dependence on the propagation delay of the carry cells. Since the propagation delay results demonstrate to be largely unaffected within the temperature range of 24°C – 39°C , we assumed this range to be relatively temperature-independent. This gave us the insight to maintain the FPGA temperature within this range to achieve stable performance for our TDC for effective use in different time critical applications.

For time critical applications, the TAC method is another important method to measure the TI and phase difference between two pulse signals. In this article, we present a novel TAC to demonstrate that a simple and very low-cost circuitry can measure the phase difference (or TI) of two pulse signals as low as 18 ps.

There is tremendous literature showing different implementation methods of TACs [21], [22], [23], [24], [25], [26], [27], [28], [29], [30]. Most TACs shown in the literature are designed and fabricated as application specific integrated circuits (ASICs). For instance, for the TAC implementation in [22], 0.35- μ m Si-Ge BiCMOS technology is used for TCSPC. The TAC in [23] is fabricated in the 180-nm standard CMOS technology for a hybrid TDC. The TAC in [24] is designed and fabricated using CMOS 0.18- μ m technology for TCSPC. Complex designs and long production times to market are the main disadvantages of TAC implementations using ASICs when compared with FPGAs. In addition, testing and design improvements are easier using FPGAs.

In this article, we propose a new method for TAC implementation using an FPGA rather than ASIC or discrete components. We use XOR gate operation based on lookup table (LUT) of low-cost XC7A100T-1CSG324C FPGA, which is fabricated using the 28-nm technology. We selected to use existing XOR gates within the FPGA to decrease the cost of the TACs. This is possible since the XOR gates are as abundant as their LUTs in the FPGA chip. Commercial XOR gates that can be used to detect picosecond-order phase shift are expensive. For instance, the cost of a commercial fast XOR gate, HMC721LP3E from Analog Devices, Inc., Wilmington, MA, USA, is around U.S. \$90 [31]. This adds an important cost when multichannel TAC is considered. For instance, the price of XOR gates for a TAC with 160 channels corresponds to an outstandingly high cost of U.S. \$14 240. This cost can be dramatically decreased by using our novel, low-cost, low-complexity TAC method. It is important to note that for every additional channel of the TAC, only one XOR gate of the FPGA and a simple, low-cost external circuitry outside of the FPGA are needed. This is crucial in building a multichannel TAC without consuming a notable chip area and without important increase in cost.

The circuitry outside of the FPGA includes a simple, low-cost RC filter and three cascaded voltage followers. These voltage followers are used behind the first order low-pass filter to decrease the settling time of the TAC output, thus increasing the sampling rate. We carried out detailed experiments showing the performance of the TAC. Temperature experiments are performed from 3°C to 93°C and linearity is shown by

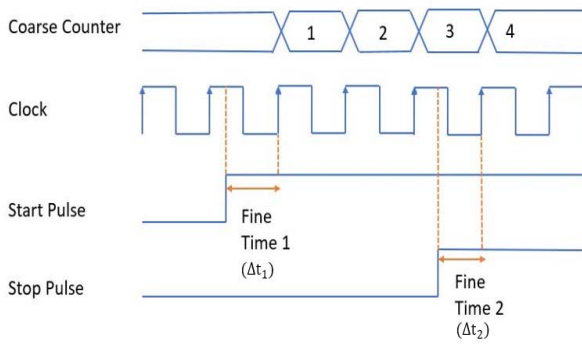


Fig. 1. Nutt interpolation method.

integral nonlinearity (INL) and differential nonlinearity (DNL) tests. Our experiments showed that at 33 °C, our TAC can detect as low as 18-ps phase increment. The DNL is shown to be between -0.39 and 0.668 LSB and the INL between -1.9 and 3.011 LSB where the LSB is 18 ps.

In addition, tests are performed for changes in the supply and input voltages. It is observed that our TAC has high resistance to PVT variations. We performed experiments up to 74 ns with 1-ns steps. The TAC is shown to have the DNL between -0.107 and 0.051 LSB and the INL between -0.299 and 0.038 LSB where the LSB is 1 ns. This shows that the designed TAC can also be used in relatively long ranges.

II. PRINCIPLE OPERATION OF THE FPGA TDC

Measuring the wide dynamic range while protecting the picosecond-order resolution is important. To maintain a resolution in the picosecond order, TDLs are used. The number of delay elements used to build a TDL is limited. Therefore, a method based on only TDLs is restricted to a hundred nanosecond measurement range at most. Thus, we used the Nutt interpolation method [32], which uses a coarse counter to increase the measurement range to the millisecond order. The measurement range depends only on the number of the counter bits of the coarse counter. In Fig. 1, the time measurement principle of the Nutt interpolation method is illustrated.

The time between the start pulse and the first rising edge of the clock is measured as the first step. As shown in Fig. 1, this measured time is called fine time 1 (Δt_1). A coarse counter starts counting with this first rising edge of the clock signal. When the stop signal arrives at the FPGA TDC, another fine time measurement takes place and measures the time between the stop pulse and the following rising edge of the clock. This measurement is called fine time 2 (Δt_2). This rising edge is counted by the coarse counter, and then, the coarse counter stops working. The TI between the start and stop pulses is calculated using the following equation:

$$TI = \Delta t_1 + T_c \times (M - 1) - \Delta t_2 \quad (1)$$

where T_c is the period of the clock signal, and M is the counted number of the rising edge of the clock signal by the coarse counter.

The structure of the proposed FPGA TDC is shown in Fig. 2. The mentioned fine time 1 is measured using carry4 delay line 1 and the D flip-flops connected to it while fine time 2 is measured by carry4 delay line 2 and its corresponding D flip-flops. The output of one measurement consists of two thermometer codes; each of them has 312 binary numbers and a 24-bit coarse counter. Each thermometer code is converted to a 9-bit binary number, and they are transferred to the computer with 24 bits of the counter via the universal asynchronous receiver and transmitter (UART). Our FPGA board uses Future Technology Devices International Ltd (FTDI) FT232HQ Hi-speed USB 2.0 slave to dual-channel UART/serial converter. This chip enables us to send our data at 12 Mbaud using the UART.

A Python code is written to calculate the TI according to (1). The 100-MHz system clock is used to synthesize both the 100-MHz fixed and phase-shifted clocks for DPS calibration and the 200-MHz clock signal for the TI and pulswidth measurements using the mixed-mode clock manager (MMCM).

There are three modes of measurements in our FPGA TDC. These are the pulswidth, the TI, and the calibration measurements. The measurement mode determines the operation before triggering the delay lines to make a time measurement, as shown in Fig. 2. The pulswidth measurement uses the same principle as the TI measurement. The only difference is that in the pulswidth measurement, the start and stop signals are given inside the FPGA according to the rising or falling edge of the incoming pulse. Calibration measurements are carried out after 10000 measurements or after a determined time. The calibration measurement is a TI measurement too. The start and stop signals of the calibration measurement are provided by the fixed and phase-shifted clocks to measure the propagation delay of the delay cells and record the attained values. The clock enable pin of the D flip-flops is controlled by the clock enable and disable circuits. The clock enable is turned to “0” after the start and stop signals trigger the delay line and the first corresponding rising edge of the fixed clock comes to transfer the output of the triggered delay cells to the output of the D flip-flops. By this way, the output of the D flip-flops is not affected by any change in the output of the delay cells. The clock enable is turned to “1” after the UART sends all the bits to the computer and the FPGA TDC is ready for a new measurement. The detailed working principle of the FPGA TDC will be explained in Sections II-A and II-B.

A. Fine-Time Measurements

Every logic gate has a propagation delay due to the propagation delay in the transistors inside them. Fast carry logic inside modern FPGAs such as Xilinx Artix-7 series are designed to make calculations as fast as possible. The propagation delay of a carry cell depends on the fabrication technology, its location inside the FPGA chip, as well as the family of the FPGA it belongs to. In our design, we use AMD Xilinx xc7a100tcs324-1 FPGA chip from the Artix-7 family. Based on our experiments, the propagation delay of 278 carry cells corresponds to a one clock period, which is 5 ns in our design at an XADC temperature of 19 °C. In addition, the mean propagation delay of one carry cell is approximately 18 ps. The

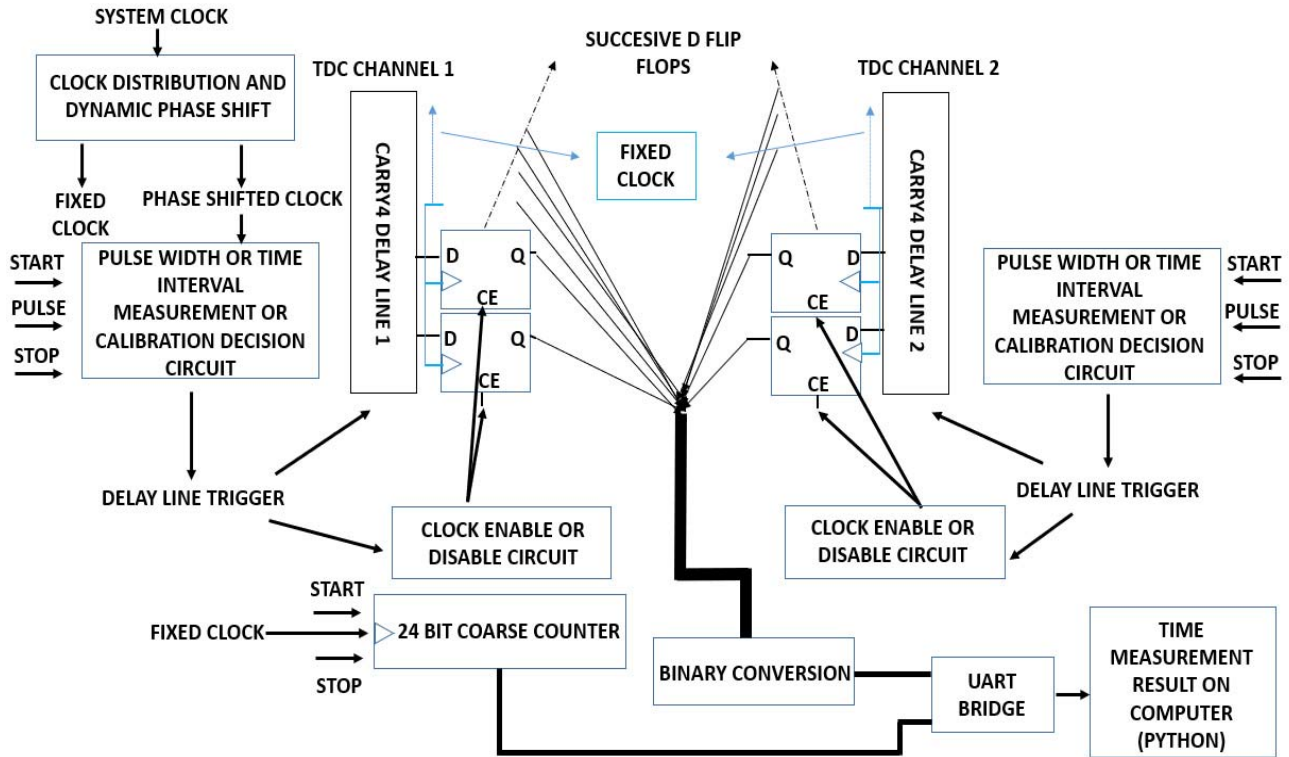


Fig. 2. General structure of the designed FPGA TDC.

maximum resolution in our TDL-based FPGA TDC depends on the propagation delay of a single carry cell since multiple TDL interpolation techniques are not applied.

In Fig. 3, the carry4 primitive available in the seven series FPGAs of AMD Xilinx is shown [33]. The logic gates designated with carry multiplexer (MUXCY) are fast multiplexers. The outputs of the multiplexers are called the carry out signals. The inputs of the first multiplexer are carry-in initialization input (CYINT) and carry cascade input (CI) signals. The CYINT signal is used to trigger the start and stop delay lines. The CI signal of the first carry4 block is logic “0,” but the other CI signals of the successive carry4 blocks are connected to the fourth carry out signal of the previous carry4 block to maintain the propagation of the trigger signal throughout all carry cells in the TDL.

Logic “1” is assigned to all the selection bits of the multiplexers, S0, S1, S2, and S3, and logic “0” is assigned to all the carry-MUX data input (DI) bits DI0, DI1, DI2, and DI3. When the start pulse triggers CYINT, the multiplexers behave just like a buffer and a logic “1” propagates through the four multiplexers in the carry block and in all the carry cells in the TDL. The outputs of the multiplexers are also connected to XOR gates designated by XORCY. The other inputs of the XOR gates are the S bits. After all the S bits are set to logic “1,” the output of the XOR gates becomes “1.” However, when the outputs of the multiplexers, which are connected to the next multiplexers and XORs, become “1” due to the propagation of the triggering signal, the output of the XOR gates becomes “0.” Thus, in addition to the carry out

signals, the output of the XOR gates can be used for fine-time measurements. We used only carry out signals in our design, but in some cases like in a tuned delay line [34], the output of the XORs can be used instead of some carry out signals as a pattern to increase linearity since all carry cells do not have exactly the same propagation delay.

In our design, we implemented two TDLs called the start and stop delay lines. The start signal triggers the start delay line while the stop signal triggers the stop delay line. When these delay lines are triggered with the start and stop signals, a logic “1” propagates throughout the delay line and gets transferred to the input of the D flip-flops. The D flip-flops are triggered by the rising edge of a clock signal, and the values at the input are copied to the output.

Clock skew is important to consider while calculating the TI result. In ideal cases, all D flip-flops’ clock signals should arrive at the clock input of the D flip-flops at the same time. However, their arrival time changes according to the position of the D flip-flop inside the FPGA chip. The net load delay (the clock signal arrival time) according to the D flip-flops is shown in Fig. 4.

Since the net load delay of neighboring D flip-flops is almost the same, we looked at eight consecutive D flip-flops. For the second delay line, the maximum and minimum net load delays are 1740 and 1707 ps, respectively. This leads to a TI calculation error of 33 ps. This error is also 33 ps for the first delay line since the maximum and minimum net load delays are 1749 and 1716 ps, respectively. In addition, there is a 9-ps offset between the two delay lines. As shown in Fig. 4, the net

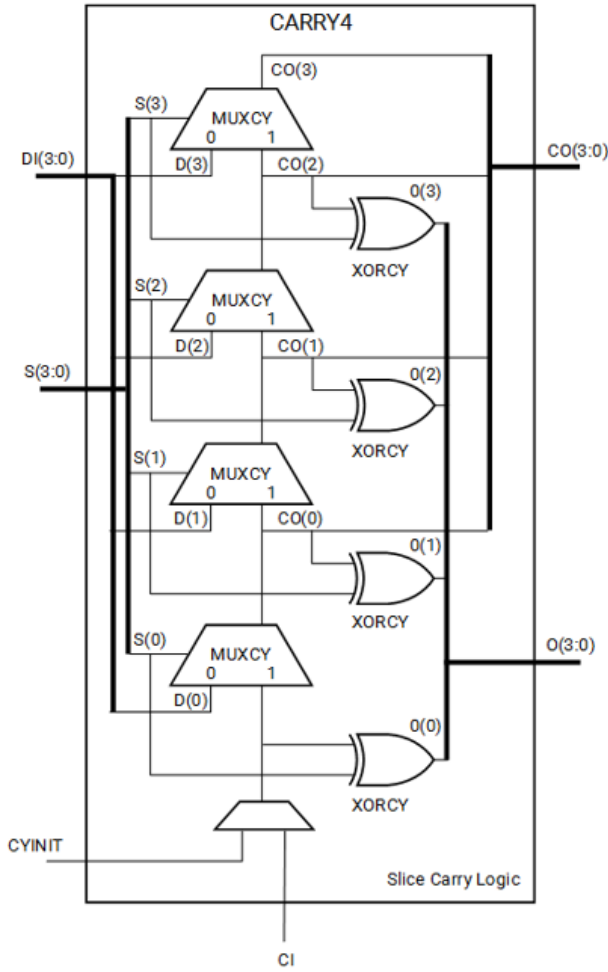


Fig. 3. Carry4 primitive of seven series AMD Xilinx FPGAs [33].

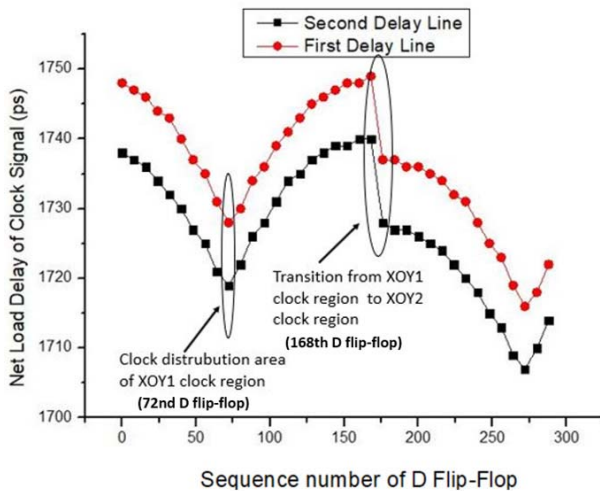


Fig. 4. Clock signal arrival time to the D flip-flops.

load delay decreases until the 72nd D flip-flop and then starts to increase suddenly. The reason behind this behavior is that the 72nd D flip-flop is so close to the clock distribution area of the clock region XOY1; hence, as the clock signal goes away up or down from that point, additional routing delay emerges.

Another important thing to consider is that after the 168th D flip-flop, there is a sudden decrease of the net load delay, as shown in Fig. 4. This is because of the clock transition region. In accordance with the implementation results of Vivado Design Suite 2019.2, the 175th D flip-flop is on the XOY1 clock region, and the 176th D flip-flop is on the XOY2 clock region. Therefore, there is a sudden change in the net load delay after the 168th D flip-flop. The clock signals are also distributed from this region to the inner areas; hence, the net load delay decreases after the 168th D flip-flop, yet it increases until then.

We implemented the sum of the bits using the or-based bubble error correction method [35] for the thermometer to binary conversion which converts the number of logic “1” to a binary number. Our difference from [35] is that we convert a 312-bit long thermometer to a 9-bit binary number, yet the method shown in this article converts a 63-bit long thermometer to a 6-bit binary number. After the binary conversion, we send the binary number of the two delay lines in addition to the coarse counter to the computer to calculate the TI.

B. Calibration Using DPS Method

So far, we have calculated the average fine time using the mean propagation delay of 18 ps for each carry cell in our FPGA chip. However, the propagation delays of all carry cells are not equal. For instance, one can be 15 ps while another one is 32 ps. Therefore, the propagation delay of every individual carry cell must be known to increase the time measurement precision.

The propagation delay of carry cells is also affected by the PVT variations. We carried out experiments to measure the effect of temperature changes on the propagation delay of carry cells. We counted the total number of carry delay cells with propagation delay corresponding to 5-ns TI which is the period of the clock signal of the FPGA TDC. We placed the FPGA board inside an ESPEC SFH-261 temperature chamber and changed the temperature from -30°C to 105°C while observing the change in the number of the carry delay cells by monitoring the temperature of the FPGA through the XADC. The 1024 measurements were carried out at approximately every 5°C temperature change of the XADC, and the average value was taken into account since the jitter in the clock of the FPGA can affect the number of the carry delay cells. It was detected that the FPGA XADC temperature is around 34°C when the surrounding is at room temperature.

Fig. 5 shows the experimental results for the total number of carry delay cells within 5-ns propagation delay versus the temperature read by the XADC. Although the two delay lines show similar behavior, the first delay line is affected much more from temperature changes below 0°C . It can be observed that the number of carry cells with the total propagation delay of 5 ns is increased up to 29°C , and then decreases for temperature changes above 39°C . Hence, our zero temperature coefficient point corresponds to this temperature region.

As shown in Fig. 5, from -30°C to 19°C , the number of carry cells increases from 230 to 280 for the first delay line

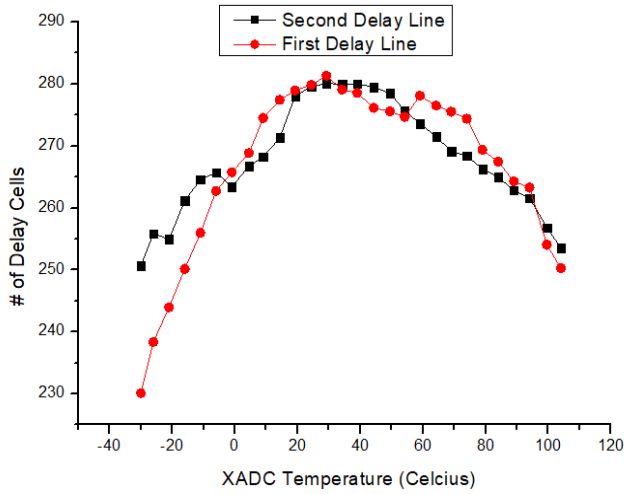


Fig. 5. Effect of temperature change observed by XADC temperature sensor on the change in the total number of carry delay cells within 5-ns propagation delay.

while it increases from 250 to 278 for the second delay line. This corresponds to a decrease in the propagation delay of one carry cell from 21.7 to 17.85 ps for the first delay line and from 20 to 17.9 ps for the second delay line. In addition, within the temperature range of 24 °C–39 °C, the mean number of carry cells for 1024 measurements is almost constant so this can be considered relatively temperature-independent within this range. Therefore, a board temperature controller can be designed to use the temperature of the XADC as a feedback to maintain the temperature within 24 °C and 39 °C. Accordingly, we can ensure the temperature-independent performance for the FPGA TDC in environments with unstable temperatures.

We conducted our calibration using the DPS feature of AMD Xilinx Artix-7 series FPGAs. Our phase-shifting resolution is 18 ps. According to [36], the DPS resolution is determined as 1/56 of the period of the voltage-controlled oscillator (VCO) of AMD Xilinx Artix-7 FPGAs. The frequency of our FPGA's VCO is 1000 MHz, which corresponds to a phase-shift resolution of 17.86 ps. This frequency is used to synthesize 100-MHz clock signals.

We have implemented a postimplementation timing simulation and experiments to test the DPS resolution of our FPGA chip and observed that the resolution is approximately 18 ps. Although a recently published article presents the DPS calibration method [18], the resolution of the phase increment is limited to 78 ps. Therefore, it is not a detailed enough calibration for our 18-ps high-resolution FPGA TDC. In our DPS calibration, a 100-MHz system clock is used to generate two 100-MHz output clocks using the MMCM. One of them has the same phase with the system clock while the other clock's phase can be changed dynamically. The fixed clock is used to trigger the D flip-flops while the variable clock is used to trigger the delay lines. By this way, a TI from 18 to 5500 ps is produced with an 18-ps resolution for calibration. Since the resolution of our FPGA TDC is 18 ps, we can calibrate almost all the delay cells which are located in the two channels.

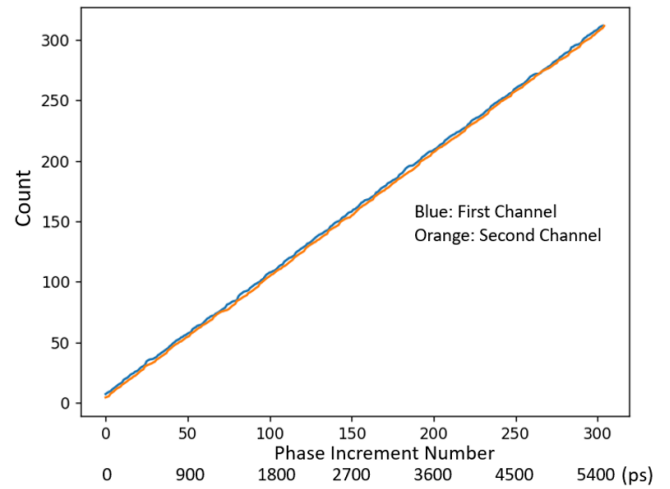


Fig. 6. Measurement results of two channels for 18-ps DPS calibration.

Although the number of effectively used carry delay cells is 278, we constructed 305 carry delay cells for test purposes. The measurement results of the first and second channels for the 18-ps DPS calibration are shown in Fig. 6.

For the calibration shown in Fig. 6, 1024 measurements are carried out for every 18-ps TI increment, and its average value is the result of the measurement for that TI. The reason for taking 1024 measurements for every phase increment is to decrease the effect of the clock jitter. In total, 312 320 measurements are done. The 312 320 measurements are carried out at 122 435 230 clock cycles corresponding to 1.2243523 s. The UART transmission time analysis is important to analyze and decrease the calibration time. Since our UART data rate is 12 Mbaud, then every bit should be sent at 83.33 ns. The clock frequency used for the UART module is 400 MHz; hence, we count 33 clock cycles to send every bit.

To decrease the usage of chip area, we implemented one thermometer to binary converter instead of two. Therefore, the 9 bits of the 312 carry delay cells of each delay line are sent in an order, not in a concurrent way. Each delay line's 9 bits are sent as 16 bits in 8-bit packets. There is one start bit, one stop bit, and no parity bit. Therefore, 8 bits are sent using 10 bits. It corresponds to 330 clock cycles which is 825 ns. We put a 30-ns blank period between two 10-bit transmission events. Thus, the time taken to send measurement results of two delay lines of the FPGA TDC for 1 of the 1024 measurements for one phase shift step is 3.390 μ s.

Another important factor affecting the dead time and sampling rate is the thermometer to binary conversion time. Although the conversion time changes depending on the thermometer value, the worst conversion time is 12.247 ns according to our postimplementation timing simulation. Therefore, we wait two clock cycles which is 20 ns for one delay line thermometer to binary conversion. There is a 10-ns delay for starting the other delay line thermometer to binary conversion, and we wait 20 ns for this delay line's thermometer to binary conversion. Therefore, the thermometer to binary conversion

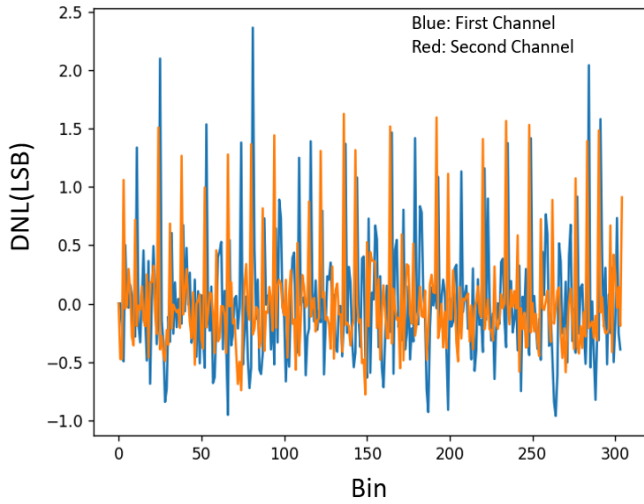


Fig. 7. DNL test.

for 1 of 1024 measurements for one phase shift step takes 50 ns.

One other significant factor affecting the total calibration time is the DPS. Before enabling the one-step phase shift increment, the signal responsible to show the phase shift should be changed from low to high. The signal to enable the dynamic phase shift should be high within only one clock cycle. After asserting this signal, the MMCM makes the `ps_done` signal high. This event shows that the phase shift between two clocks is increased by one step, which is 18 ps in our design. There are 13 clock cycles corresponding to 130-ns TI between the phase shift enable pulse and `ps_done` signal. Therefore, the time needed for one phase shift increment is 140 ns including the phase shift increment signal.

We used the DNL test to measure the deviation of propagation delay of carry delay cells from their mean propagation delay of 18 ps. The DNL is shown in the LSBs. The LSB for our FPGA TDC is 18 ps. DNL of the first channel is between -0.95 and 2.37 LSB while DNL of the second channel is between -0.78 and 1.63 LSB as shown in Fig. 7.

The accumulated DNL at a bin number gives the INL. It shows the total deviation from the ideal fine-time measurement at that bin number. The INL of the first channel is between -2.46 and 3.65 LSB while the DNL of the second channel is between -1.96 and 3.86 LSB as shown in Fig. 8.

III. EXPERIMENTAL RESULTS OF THE FPGA TDC

In our experiments, we used Keysight 350-MHz universal frequency counter/timer 53230A and Keysight 33600A series trueform waveform generator [37]. The TI or pulse signals that will be used to test the FPGA TDC must be uncorrelated to the clock of the FPGA to be able to get better, real life test results which include the effects of the environment and internal clock noise.

We produced pulse signals with different pulse widths using the waveform generator and measured these pulses using the frequency counter/timer to ensure that the produced pulsewidth time is correct. The frequency counter/timer has a single-shot

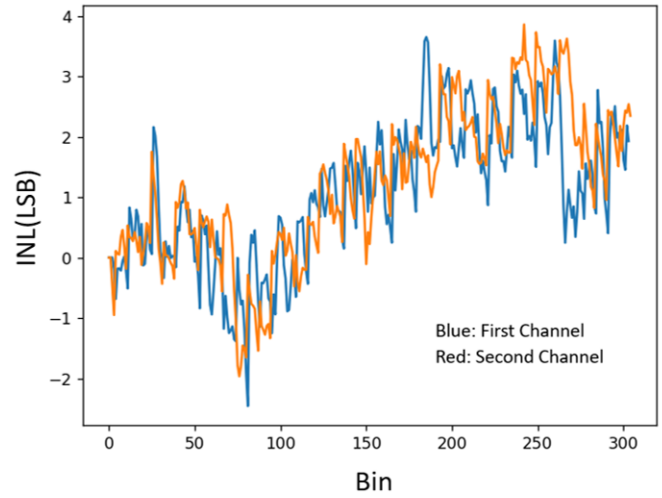


Fig. 8. INL test.

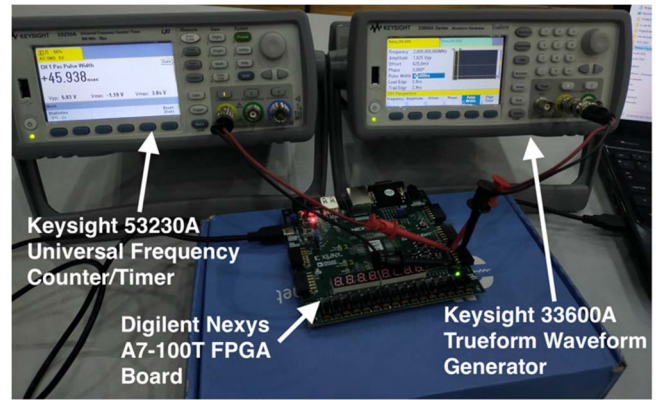


Fig. 9. Test setup of the FPGA TDC.

precision of 20 ps. We tested different BNC connectors, cables, and probes to get the best signal with the lowest jitter. The experimental setup for the FPGA TDC is shown in Fig. 9.

The BNC connector and cable used in the test setup provided the clearest signal with standard deviation from 18 to 30 ps. The FPGA TDC requires a start signal and a stop signal to measure the TI between them, but a pulse is generated as one signal. Our waveform generator could not provide a phase shift below 1 ns between two channels. Therefore, to provide the start and stop signals, we wrote an additional VHDL code. The waveform generator generates a pulse signal, and we convert this pulse signal into two pulses (start and stop pulses). When the rising edge of the input signal occurs, the internal start signal is given. In the same manner, the stop signal is produced after the falling edge of the input signal to test the FPGA TDC. By this way, the pulsewidth time of the generated pulse via the waveform generator is converted to TI to test the FPGA TDC.

As a first experiment, we generated pulse signals with pulsewidth from 10 to 200 ns with 5-ns steps. We waited to observe the mean to be approximately constant, and then, we attained the single-shot precision of the FPGA TDC using standard deviation information of the histogram of the

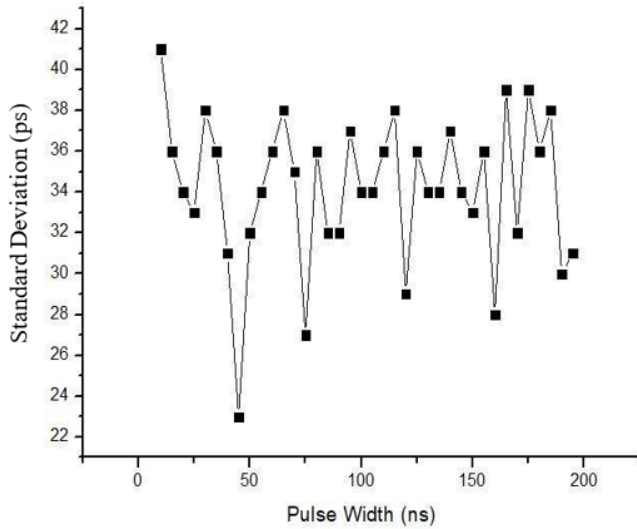


Fig. 10. FPGA TDC single-shot precision test up to 200 ns.

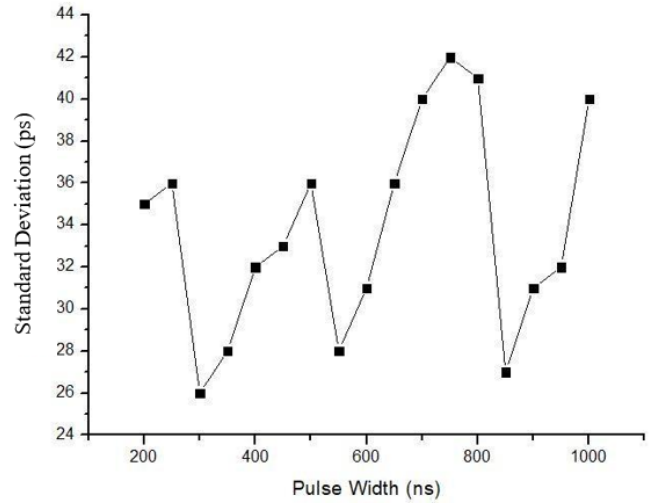
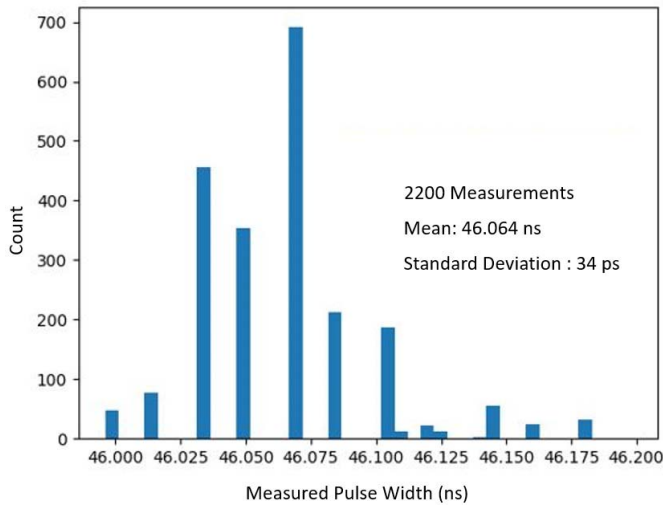
Fig. 12. FPGA TDC single-shot precision test up to 1 μ s.

Fig. 11. Histogram of 46.05-ns pulsewidth for 2200 measurements.

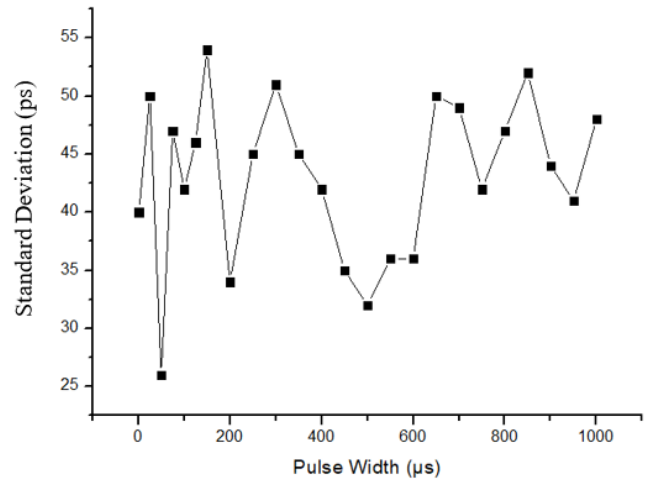


Fig. 13. FPGA TDC single-shot precision test up to 1 ms.

measurement results. The obtained standard deviation up to 200 ns is shown in Fig. 10. The mean of the standard deviation in the range of 10–200 ns is 34.1 ps. The best attained standard deviation is 23 ps while the worst standard deviation is 41 ps.

An example histogram with 2200 measurements is shown in Fig. 11. A 2-MHz pulse signal with 46.05-ns pulsewidth is generated by the waveform generator, and 2200 pulsewidth measurements are done and 34-ps precision is achieved.

The pulsewidth measurement to test the performance of the FPGA TDC for the range from 200 ns to 1 μ s was carried out. Measurements were done with 50-ns steps in the same method with the previous single-shot measurements shown in Fig. 10. The attained standard deviation plot is shown in Fig. 12. The mean standard deviation is 33.7 ps. The best attained standard deviation is 26 ps while the worst standard deviation is 42 ps.

The same single-shot precision test which is carried out for the range from 1 to 125 μ s with 25- μ s steps and for the range from 150 μ s to 1 ms with 50- μ s steps is shown in Fig. 13.

The best attained standard deviation is 26 ps while the worst is 54 ps and the mean is 42.3 ps. This result is important and shows that our FPGA TDC's single-shot precision does not change much from 10 ns to 1 ms so it is stable. Achieving this stable precision for such a wide dynamic range with a very low-cost FPGA is important to decrease the cost of the TDCs. In addition, this is achieved using only the clock of the FPGA board itself so low complexity is also accomplished. The clock of the Nexys A7-100T is supplied by DSC1033CC1-100.0000T from Microchip Technology, Chandler, AZ, USA, [38]. This is a 100-MHz 3.3-V fixed-frequency MEMS-based low-power CMOS oscillator.

Our aim is to achieve high precision in a long TI range with a low-cost general-purpose FPGA. Szplet *et al.* [39] and Zhang *et al.* [40] also show high precision in the long TI range. However, in [39], external low jitter frequency synthesizer which synthesizes a 500-MHz signal using a 10-MHz reference signal of a TCXO (0.5-ppm stability) and a rubidium

atom-based generator FS725 are used. This rubidium generator and TCXO add an important cost to the FPGA TDC while our aim is to design the FPGA TDC using a very low-cost Nexys-A7-100T FPGA board. In addition, rubidium generators are applied to the arbitrary waveform generators in [39]. In our experiments using Keysight 53230A, we observed that the standard deviation of our Keysight 33600A waveform generator is getting worse up to 30 ps when the range gets closer to milliseconds. Thus, our waveform generator adds to the majority of the jitter when we consider our worst standard deviation of 54 ps in the range of 1 ms. In [40], a 100-MHz oven controlled crystal oscillator (OCXO) is used for the input clock of the FPGA, and it increases the cost and complexity of the board since it requires a stable temperature to operate. In addition, the cost of the used Virtex-5 FPGA chip in [40] is at least 3–4 times larger than our low-cost Artix-7 FPGA chip. In [41], an expensive Kintex-7 FPGA is used for high precision in a wide dynamic range. We managed through using only a low-cost FPGA board to achieve sufficiently high and stable single-shot precision over a wide dynamic range.

Using filters like moving average for the finite impulse response (FIR) increases the usage area in the FPGA chip and decreases the sampling rate a lot. For instance, in [42], a 128-tap FIR filter using a moving average is implemented to increase the rms resolution from 70 to 8 ps. However, such an increase in rms resolution decreases the sampling rate to 1 kHz which is not sufficient for many applications. In addition, the used FIR filter causes a high usage in the FPGA chip area. In [42], 2962 LUTs and 4157 flip-flops per channel are used. The precision can be increased with a multiple TDL like in [43]. In [43], four TDLs are merged to attain a 3.9-ps TI rms precision. However, the resources' usage is 6258 flip-flops and 2433 LUTs, which is significantly higher than our work. This usage of resources is an obstacle in implementing TDCs with low-cost FPGAs. In [44], an FPGA TDC is designed with a 69-ps rms resolution and a 59-ps single-shot precision using Zynq Ultrascale + MPSoC for the TDC architecture based on a gray code oscillator. In this work, each TDC channel requires seven LUTs and 20 registers while the complete TDC IP requires 482 LUTs and 409 registers. However, when all the IPs are included as a complete TDC system, 2064 LUTs and 2696 registers are used. In [45], an ultrascale FPGA is used to design a 128-channel TDC for LiDAR applications with a resolution from 51.28 to 105.26 ps and with a precision from 15.89 to 26.32 ps. The TDC is tested up to 1000 ns. One channel consumes 660 LUTs, 1100 registers, and 2.5 block random access memory (BRAM). In [46], a multichannel, low nonlinearity TDC is shown, in which the LSB of 10.5 ps is achieved by Virtex 7 and the resources' usage is 1145 LUTs and 1916 flip-flops. Instead, we consume 348 LUTs and 818 flip-flops if we implement thermometer to binary conversion outside of the FPGA to build our FPGA TDC with two TDLs. This decreases the clock jitter effect which increases the single-shot precision in the millisecond range. When the thermometer to binary conversion is implemented by VHDL, then our FPGA TDC consumes 724 LUTs and 824 flip-flops.

In Fig. 14, a histogram of 2190 measurements of a pulse signal with a frequency of 200 Hz and 300- μ s pulsewidth

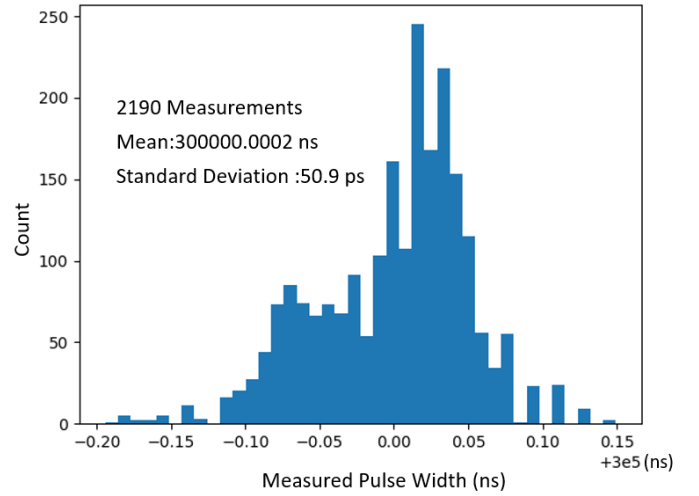


Fig. 14. Histogram of a 300- μ s pulsewidth for 2190 measurements.

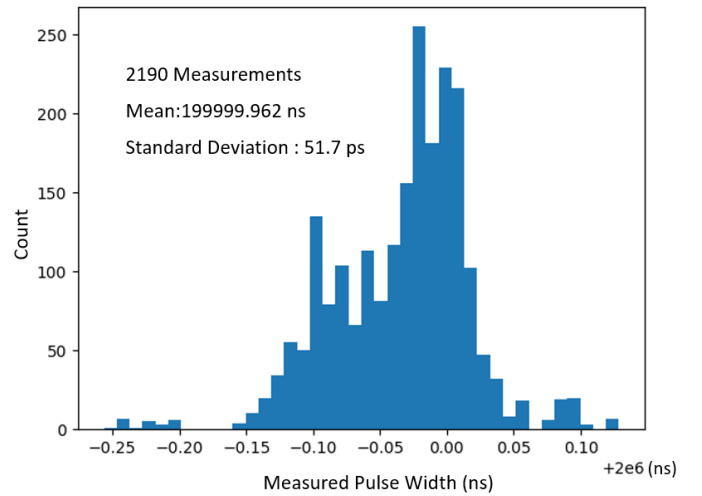


Fig. 15. Histogram of a 2-ms pulsewidth for 2190 measurements.

is shown. This pulse signal is again generated with Keysight 33600A. The attained standard deviation is 50.9 ps. In Fig. 15, a histogram of 2190 measurements of a pulse signal with a frequency of 20 Hz and 2-ms pulsewidth is shown. The obtained standard deviation result is 51.7 ps. This result shows that we can measure the TI in the millisecond range with a precision of 51.7 ps using a low-cost FPGA board.

IV. PRINCIPLE OPERATION OF THE TAC

The reason we call our circuit the TAC is that it converts the phase difference, in other words, the time difference of two pulses to a dc voltage value. Our TAC includes only one XOR gate and a simple circuitry. We used a simple XOR gate which is controlled by an LUT of one slice in the FPGA fabric. One slice consists of four six-input LUTs, three 2×1 multiplexers, one carry4 block, and eight flip-flops. There are 8150 cells in our low-cost AMD Xilinx XC7A100T-1CSG324C FPGA. One channel TAC uses only one cell and one LUT of this cell.

The main application area for using the TAC is the continuous wave LiDAR (CW-LiDAR). A low jitter square wave

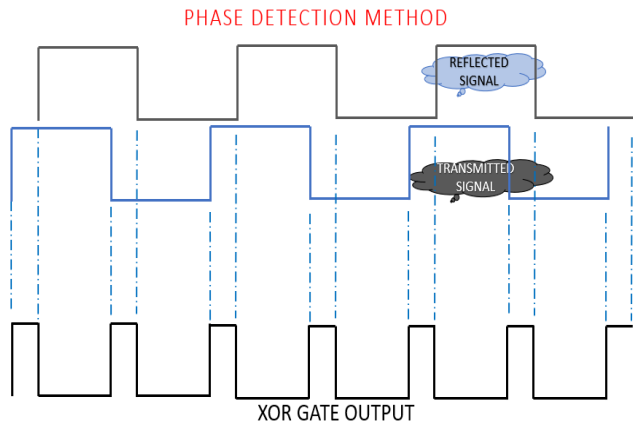


Fig. 16. Example demonstration of phase shift detection between transmitted and received signals of a CW-LiDAR using an XOR gate.

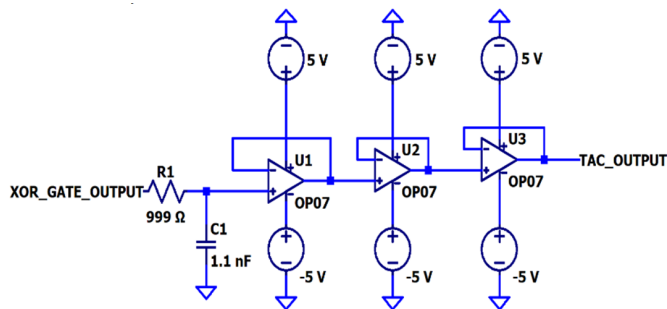


Fig. 17. Designed TAC circuitry.

signal is generated by an FPGA, and it drives the driver circuit of a laser diode to transmit the light. The transmitted light travels in the air and gets reflected by an object, and then, it is detected by the photodiodes of the receiver of the CW-LiDAR. Low photodiode current is amplified using a fast trans-impedance amplifier circuit and is converted to a voltage signal. This voltage signal is compared to a threshold voltage by a comparator with a good hysteresis or Schmitt trigger circuit to make the signal a square wave again. An example demonstration of the transmitted and reflected signals is shown in Fig. 16. There is a phase shift between the two signals since the transmitted signal returns after a time which depends on the distance of the object. As a first step to measure the distance, the XOR gate of the FPGA is used to detect this phase shift. The period of the signal at the output of the XOR gate is half of the transmitted and reflected signals. The pulsewidth of the output signal of the XOR gate is equal to the phase shift between the transmitted and received signals, as can be seen in Fig. 16. Therefore, the duty cycle of the square wave signal depends on the phase shift and the frequency of the transmitted signal. Since the frequency of the transmitted signal is constant, the duty cycle information gives us the phase shift between the transmitted and the reflected signals accordingly. Therefore, we can measure the duty cycle to determine the phase shift and we can use this phase shift value to calculate the distance of the object.

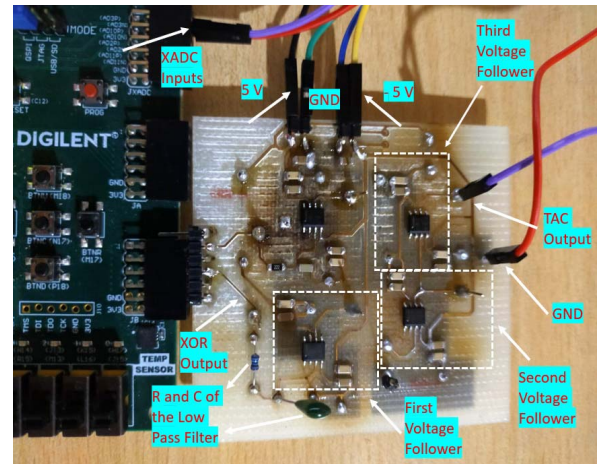


Fig. 18. PCB of the TAC and its connection to the FPGA.

To measure the duty cycle of the output signal of the XOR gate to determine the phase shift, we designed a simple and low-cost TAC. Our simple TAC circuit schematic is shown in Fig. 17. This circuitry can be used to determine the duty cycle and the pulsewidth of a pulse or a square wave signal in a very accurate way.

The output of the TAC should have minimum ripples. As a first step, the first-order simple low-pass filter is used to convert the XOR gate output to a dc voltage with minimum ripples. Decreasing the settling time increases the ripples of the filter output. If the peak-to-peak voltage (V_{pp}) of the TAC output increases, the XADC of the FPGA should average more samples to find the mean value; hence, it decreases the sampling rate of the TAC. Our aim is to get a maximum 3 mV of V_{pp} ripple at the TAC's output. The minimum RC time constant of the first-order low-pass filter which provides 3-mV V_{pp} output is $22 \mu s$ using a 2.2-k Ω resistor and a 10-nF capacitor. Since a few millivolts error corresponds to approximately 50-ps error, then, we should sample the TAC's output where it settles to 99.5% of its final value as a worst-case scenario. The settling time of the TAC's output is approximately $240 \mu s$ when using a 2.2-k Ω resistor and a 22-nF capacitor in the low-pass filter and when the input of the TAC has two 2-MHz pulse signals with 50-ns phase difference. The settling time of $240 \mu s$ limits the sampling frequency to 4.166 kHz. Therefore, we designed the simple and low-cost circuit shown in Fig. 17 to decrease the settling time and thus increase the sampling frequency. The designed TAC circuit's printed circuit board (PCB) can be seen in Fig. 18.

The settling time of the designed TAC is $25 \mu s$ when two 2-MHz pulse signals with 50-ns phase difference are the inputs of the TAC. Therefore, we significantly reduced the settling time of the TAC, this increasing the sampling rate to 40 kHz. In addition, an abrupt change from 0 to 50-ns phase difference is considered; hence, it can be named as the worst-case settling time since a lower change in the phase difference such as from 10 to 12 ns will require much lower settling time. Three cascaded voltage followers are used behind the first-order low-pass filter. The resistance and capacitance values of

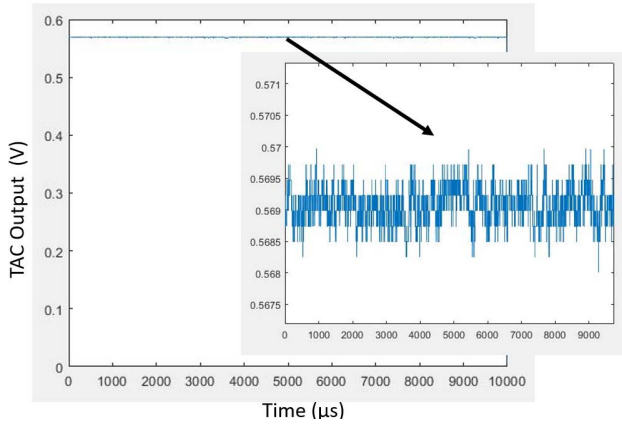


Fig. 19. TAC output sampled within 10 ms using XADC of the FPGA.

the low-pass filter are $999\ \Omega$ and $1.1\ \text{nF}$, respectively. The output of the XOR gate of the FPGA chip is connected to the low-pass filter as shown in Fig. 18. The TAC circuit converts the output of the XOR gate to a dc voltage proportional to its duty cycle. By this way, the phase shift between the two input signals is measured. The TAC output is taken from the output of the third voltage follower and connected to the differential inputs of the FPGA's XADC along with its GND pin as can be seen in Fig. 18.

V. EXPERIMENTAL RESULTS OF THE TAC

In Fig. 19, 10 000 samples of the TAC's output, which are collected within 10 ms with $1\text{-}\mu\text{s}$ steps using the 12-bit 1-MS/s differential input ADC of the FPGA, are shown to observe the noise and ripples in a detailed way. Even if the low-pass filter output has 67-mV Vpp ripples, the voltage followers decrease the ripples gradually and there is only 1.8-mV Vpp noise and ripple at the output. The operational amplifiers of the voltage followers are the low-cost OP07C model from Texas Instruments, Dallas, TX, USA.

We carried out experiments to determine how sensitive our TAC can measure the phase shift. The first experiment is done to test the relatively long-range measurement. We used the waveform generator to produce two pulse signals with a variable phase shift between them. The minimum phase shift step that the waveform generator can provide is $1\ \text{ns}$. A 2-MHz , 3.3-Vpp pulse signal with 250-ns pulsewidth is generated by channel 1 and channel 2 of the waveform generator. Channels 1 and 2 of the waveform generator are connected to the two input pins of the FPGA board. These 2 input pins are connected to the inputs of the XOR gate which are the inputs of the TAC. The output of the XOR gate is connected to the TAC, as shown in Fig. 17. The output of the TAC is connected to the XADC of the FPGA to convert the analog output voltage of the TAC to 12-bit digital values. The 12-bit digital values are sent to the computer via the UART. A code written in Python is used to convert these digital values back to analog voltage values by dividing them by 4096. The phase shift is changed from 0 to $74\ \text{ns}$ with 1-ns steps. The measurement results are shown in Fig. 20. The measurements are carried out with $\pm 5\text{-V}$ supply voltage

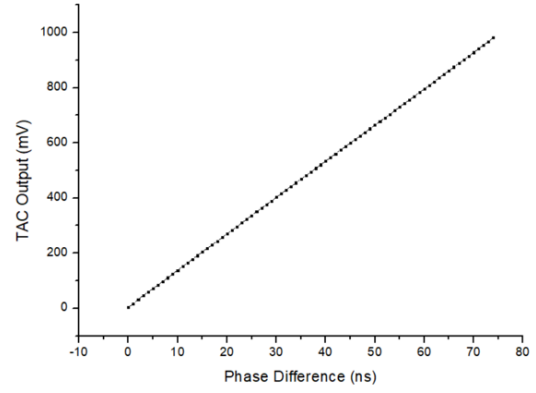


Fig. 20. Experimental results of the designed TAC up to $74\ \text{ns}$ with 1-ns steps.

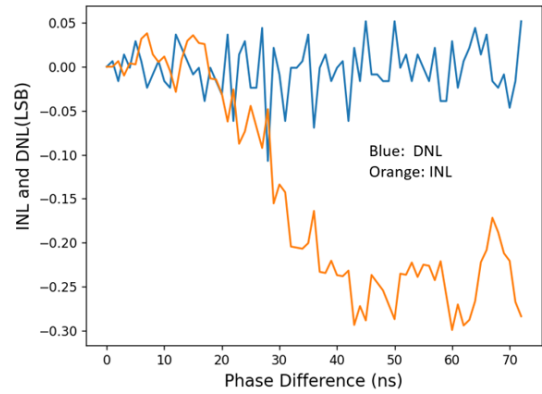


Fig. 21. INL and DNL tests for the long-range TAC experiment.

and at $33\ ^\circ\text{C}$ XADC temperature. A 1-ns phase increment corresponds to an increase in the TAC's output's mean value by $13.216\ \text{mV}$.

The TAC results in Fig. 20 show a highly linear behavior. To observe the linearity in a more detailed way, the DNL and INL tests are carried out and the results are shown in Fig. 21. The DNL is between -0.107 and $0.051\ \text{LSB}$ while the INL is between -0.299 and $0.038\ \text{LSB}$ where the LSB is $1\ \text{ns}$. Therefore, it can be deduced that the TAC shows a very linear behavior for a relatively long range which is $74\ \text{ns}$ in this experiment. Since the maximum measurable voltage of the FPGA XADC is $1000\ \text{mV}$ and $74\ \text{ns}$ corresponds to $981\ \text{mV}$, we did not increment the phase difference anymore.

We carried out another experiment to see the behavior of our TAC for a phase increment step below $1\ \text{ns}$. In this experiment, we used the DPS method that we used before to calibrate the FPGA TDC. We used two 100-MHz clocks for calibration and MMCM to synthesize these clocks by using a 1000-MHz VCO. The resolution of the phase increment at this VCO frequency is approximately $18\ \text{ps}$ according to our experiments with Keysight 53230A timer and simulation results. We changed the phase shift between two 100-MHz clocks with 18-ps phase increment steps. We produced two pulse signals called the start and stop signals using these two clocks. Their periods and duty cycles are the same while the phase between them is a variable. The period is $160\ \text{ns}$

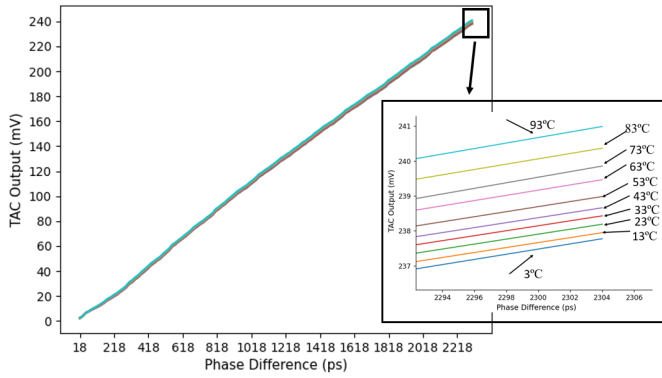


Fig. 22. TAC test with 18-ps phase increment steps for temperature change from 3 °C to 93 °C with 10 °C steps.

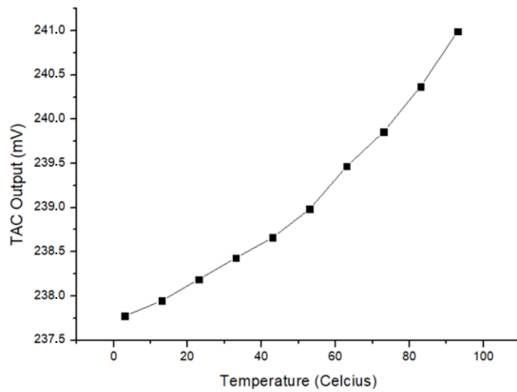


Fig. 23. Temperature effect on TAC measuring the 2304-ps phase shift.

while the duty cycle is 50%. Since the stop signal is produced by a dynamically phase-shifted clock and the start signal is produced by a fixed clock, then, the phase shift between the start and the stop signal can be changed dynamically. The phase shift resolution is 18 ps as in the DPS calibration of the FPGA TDC. The TAC's output is measured by the XADC of the FPGA.

We also carried out this experiment at ten different temperatures from 3 °C to 93 °C to observe the effect of temperature change on the performance of the TAC. We placed the TAC circuit inside the ESPEC SFH-261 temperature chamber, and the temperature information is taken from the XADC of the FPGA. The experimental results are shown in Fig. 22. We achieved a mean dc voltage change of 1.858 mV as a result of a phase shift of 18 ps at 33 °C.

As can be observed in Fig. 22, the temperature effect increases as the phase difference increases and the maximum TAC output voltage change is at 2304-ps phase difference. Therefore, in Fig. 23, we plot the TAC output for a phase difference of 2304 ps according to the temperature change to see the effect of temperature change in a more detailed way. For a large temperature change from 3 °C to 93 °C, the TAC output changes by only 3.3 mV. Therefore, we can say that even a 90 °C temperature change almost does not affect the performance of the TAC. These are promising results showing

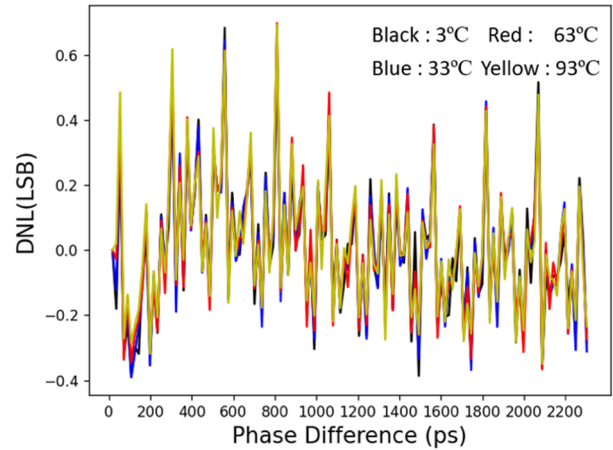


Fig. 24. DNL test for the 18-ps resolution phase shift experiment.

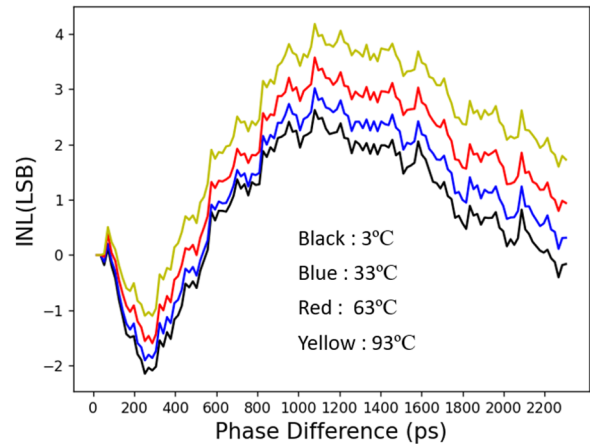


Fig. 25. INL test for the 18-ps resolution phase shift experiment.

the efficiency in measurements using our low-cost TAC with minimal complexity.

To measure the linearity of the TAC, INL and DNL tests were performed. The results of the DNL test are shown in Fig. 24. We noticed that the XADC temperature is around 33 °C at room temperature. A phase increment equal to the LSB, which is 18 ps, corresponds to an increase in the mean value of the TAC's output by 1.858 mV at 33 °C. The DNL test is carried out at 3 °C, 33 °C, 63 °C, and 93 °C. The maximum and minimum DNL values for 3 °C are 0.682 and −0.389 LSB, respectively. For the 33 °C temperature, the minimum DNL is −0.39 LSB while the maximum DNL is 0.668 LSB. For the 63 °C temperature, the minimum DNL is −0.366 LSB while the maximum DNL is 0.697 LSB. The maximum and minimum DNL values for 93 °C are 0.693 and −0.349 LSB, respectively.

The results of the INL test are shown in Fig. 25. The maximum and minimum INL values for 3 °C are 2.618 and −2.143 LSB, respectively. For the 33 °C temperature, the minimum INL is −1.9 LSB while the maximum INL is 3.011 LSB. For the 63 °C temperature, the minimum DNL is −1.591 LSB while the maximum DNL is 3.568 LSB. The

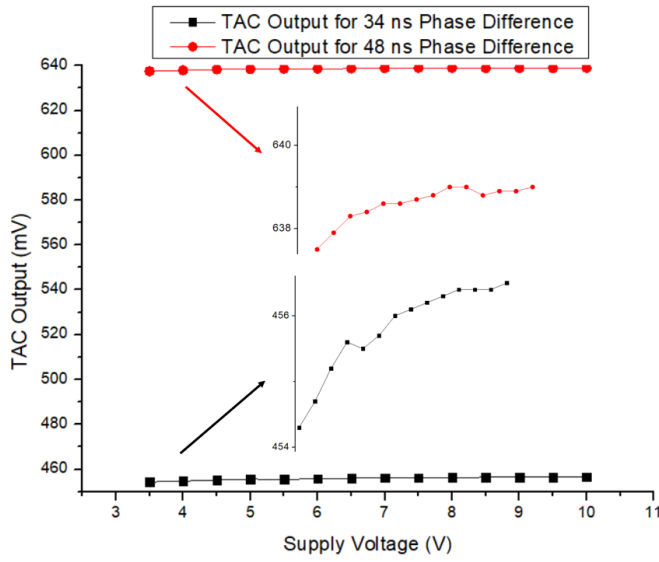


Fig. 26. Supply voltage effect on the performance of the TAC.

maximum and minimum DNL values for 93 °C are 4.173 and -1.097 LSB, respectively. DNL and INL tests are performed using ± 5 -V supply voltage. As shown in Fig. 25, the increase in the INL values for different temperature values follows a similar pattern.

Such DNL and INL results show our TAC's high resistance to temperature changes and its linear behavior even when we increase the phase shift by 18 ps. We use these DNL and INL values to calibrate our TAC to get high-precision phase difference measurements.

In addition to the temperature measurements, the effect of voltage variations on the performance of the TAC should also be investigated for use in our calibration accordingly. We did performance measurements for two different voltage variations. One is carried out by changing the supply voltage and the other is carried out by changing the amplitude of the input voltage. Since it is difficult to measure the effect of the voltage change on every phase increment value, two points are selected, 34 and 48 ns, from our long-range TAC experiment. In the first experiment, the supply voltage of the TAC is changed from ± 3.5 to ± 10 V with 0.5-V steps and the results of the experiment are shown in Fig. 26. For the 34-ns phase difference, the change in supply voltage changes the TAC output voltage to a maximum value of 2.2 mV. This value is 1.5 mV for the 48-ns phase difference. Therefore, the supply voltage change has almost no effect on the TAC performance.

Our second experiment for investigating the effect of voltage variation on the TAC performance is done through changing the amplitude of the input voltage. The results are shown in Fig. 27. The expected input voltage has 0 V as the low voltage value and 3.3 V for the high voltage value. In real life, the comparator outputs in LiDAR systems should give the output signal with at most 0.1-V deviation. However, we carried out an experiment from 1.5 to 3.3 V with 0.1-V steps. From 2 to 3.3 V, the TAC's output for 34-ns phase

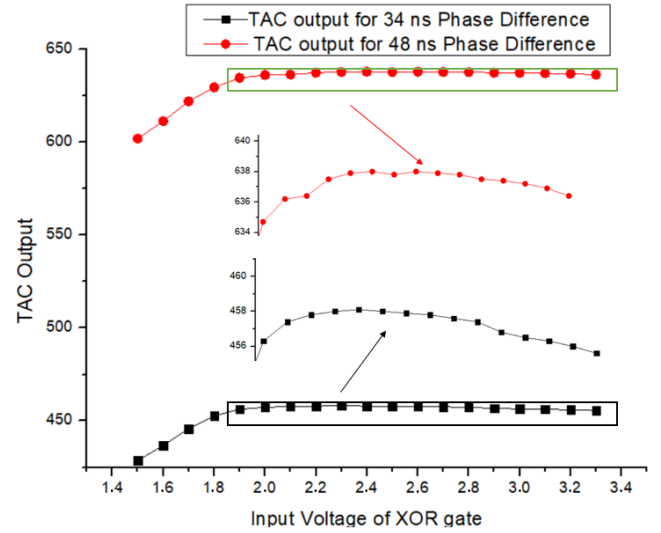


Fig. 27. Input voltage of the XOR gate effect on the performance of the TAC.

difference changes by 2.5 mV and the TAC's output for 48-ns phase difference changes by 1.8 mV. This shows that the TAC is not sensitive to the input voltage change of the XOR gate.

In addition to the input-voltage variation measurements, an important thing to consider in TAC design is the net load delay of the two inputs of the XOR gate. The XOR gate should be placed in the FPGA fabric such that the two incoming signals' routing delays, in other words the net load delays, be equal to each other just like in differential high-speed double data rate fourth generation (DDR4) memory PCB traces. By this way, incoming signals to the XOR gate will arrive at the same time and there will be no additional phase offset. We tried different locations for the XOR gate in the FPGA chip by looking at the implementation results.

VI. CONCLUSION

In this article, we presented a low-cost, wide dynamic range, high-resolution, and high-precision FPGA TDC. We also showed a novel high-resolution TAC employing the XOR gate in the FPGA along with a simple low-cost circuitry. Our designed TDC and TAC are low in cost and complexity making them desirable for LiDAR and time critical applications.

Our FPGA TDC design has a wide dynamic range from picoseconds up to milliseconds and a high single-shot precision using only a low-cost general purpose FPGA board without needing any complex and/or expensive external circuitry or instrumentations. Such a wide dynamic range has been validated through our experimental results. The mean standard deviation up to 1 μ s is shown to be 33.9 ps while the mean standard deviation is 41 ps from 1- μ s to 1-ms measurements. We also tested the linearity of our TDC through the DNL and INL tests. Our performance measurements allowed us to identify the ideal range of temperature to operate the TDC within. This range was found to be 24 °C–39 °C and is considered a temperature-insensitive region. We also presented a DPS calibration with a high resolution, as low as 18 ps, to effectively calibrate our FPGA TDC.

We also demonstrated a novel, high-resolution TAC that is low in complexity and cost using an XOR gate and XADC within an inexpensive Nexys-A7-100T board. We decreased the settling time of the TAC circuit from 240 to 25 μ s which increased the sampling rate. In addition, we carried out detailed PVT variation experiments to test the performance of the TAC. Temperature experiments were performed from 3 °C to 93 °C, and the TAC showed to be highly immune to temperature variations. Furthermore, we observed linearity from performing INL and DNL tests at different temperatures. Our experiments showed that our TAC can detect as low as 18-ps phase increment and have DNL between -0.39 and 0.668 LSB and INL between -1.9 and 3.011 LSB. Supply and input voltage variation tests were also performed. It was concluded that our TAC has high resistance to voltage variations. We performed experiments up to 74 ns with 1-ns steps and the TAC had the DNL between -0.107 and 0.051 LSB while the INL between -0.299 and 0.038 LSB. This shows that the designed TAC can be used in relatively long ranges. It is valuable to note that, every additional channel in the TAC requires only one XOR gate of the FPGA and a simple, low-cost external circuitry outside the FPGA which consists of a low-pass filter and three-stage voltage followers.

LiDAR systems are gaining more attention and popularity in wide application areas from remote sensing and autonomous vehicles to topographic mapping. However, cost is one of the main concerns which need to be addressed. We believe that our low-cost low-complexity TDC and TAC can be used for TOF measurements in LiDAR systems to reduce the overall cost.

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